LAB PROGRAM 1
a)FCFS

```
#include <stdio.h>
int main()
{
    int n, i, j;
    int pid[10], at[10], bt[10];
    int ct[10], tat[10], wt[10];
    float avg_wt = 0, avg_tat = 0;

    printf("Enter the number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        pid[i] = i + 1;
        printf("Enter arrival time and burst time for process %d: ",pid[i]);
        scanf("%d %d",&at[i],&bt[i]);
    }

    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(at[i] > at[j])
            {
                int temp;

                temp = at[i];
                at[i] = at[j];
                at[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;

                temp = pid[i];
                pid[i] = pid[j];
                pid[j] = temp;
            }
        }
    }
```

```c
    ct[0] = at[0] + bt[0];

    for(i=1;i<n;i++)
    {
        if(ct[i-1] < at[i])
            ct[i] = at[i] + bt[i];
        else
            ct[i] = ct[i-1] + bt[i];
    }

    for(i=0;i<n;i++)
    {
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    avg_wt /= n;
    avg_tat /= n;

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
    {
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
    }

    printf("\nAverage Waiting Time: %.2f",avg_wt);
    printf("\nAverage Turnaround Time: %.2f\n",avg_tat);

    return 0;
}
```

```
Enter the number of processes: 4
Enter arrival time and burst time for process 1: 1 2
Enter arrival time and burst time for process 2: 3 4
Enter arrival time and burst time for process 3: 5 6
Enter arrival time and burst time for process 4: 7 8

PID     AT      BT      CT      TAT     WT
1       1       2       3       2       0
2       3       4       7       4       0
3       5       6       13      8       2
4       7       8       21      14      6

Average Waiting Time: 2.00
Average Turnaround Time: 7.00

Process returned 0 (0x0)   execution time : 13.333 s
Press any key to continue.
```

b)SJF

```
#include <stdio.h>
#include <limits.h>

int main()
{
    int n, i, time = 0, completed = 0;
    int pid[10], at[10], bt[10];
    int ct[10], tat[10], wt[10];
    int visited[10] = {0};
    float avg_wt = 0, avg_tat = 0;

    printf("Enter the number of processes: ");
    scanf("%d",&n);
```

```
for(i=0;i<n;i++)
{
    pid[i] = i+1;
    printf("Enter arrival time and burst time for process %d: ",pid[i]);
    scanf("%d %d",&at[i],&bt[i]);
}

while(completed < n)
{
    int min = INT_MAX, pos = -1;

    for(i=0;i<n;i++)
    {
        if(at[i] <= time && visited[i]==0 && bt[i] < min)
        {
            min = bt[i];
            pos = i;
        }
    }

    if(pos == -1)
    {
        time++;
    }
    else
    {
        time += bt[pos];
        ct[pos] = time;
        visited[pos] = 1;
        completed++;
    }
}

for(i=0;i<n;i++)
{
    tat[i] = ct[i] - at[i];
    wt[i] = tat[i] - bt[i];

    avg_wt += wt[i];
    avg_tat += tat[i];
}

avg_wt /= n;
avg_tat /= n;
```

```
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
    {
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
    }

    printf("\nAverage Waiting Time: %.2f",avg_wt);
    printf("\nAverage Turnaround Time: %.2f\n",avg_tat);

    return 0;
}
```

```
Enter the number of processes: 4
Enter arrival time and burst time for process 1: 1 2
Enter arrival time and burst time for process 2: 3 4
Enter arrival time and burst time for process 3: 5 6
Enter arrival time and burst time for process 4: 7 8

PID     AT      BT      CT      TAT     WT
1       1       2       3       2       0
2       3       4       7       4       0
3       5       6       13      8       2
4       7       8       21      14      6

Average Waiting Time: 2.00
Average Turnaround Time: 7.00

Process returned 0 (0x0)   execution time : 18.196 s
Press any key to continue.
```

c) SRTF

```
#include <stdio.h>
#include <limits.h>

int main()
```

```
{
    int n, i, time = 0, completed = 0;
    int pid[10], at[10], bt[10], rt[10];
    int ct[10], tat[10], wt[10];
    float avg_wt = 0, avg_tat = 0;

    printf("Enter the number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        pid[i] = i+1;
        printf("Enter arrival time and burst time for process %d: ",pid[i]);
        scanf("%d %d",&at[i],&bt[i]);
        rt[i] = bt[i];
    }

    while(completed != n)
    {
        int min = INT_MAX, pos = -1;

        for(i=0;i<n;i++)
        {
            if(at[i] <= time && rt[i] > 0 && rt[i] < min)
            {
                min = rt[i];
                pos = i;
            }
        }

        if(pos == -1)
        {
            time++;
        }
        else
        {
            rt[pos]--;
            time++;

            if(rt[pos] == 0)
            {
                completed++;
                ct[pos] = time;
            }
```

```
        }
    }

    for(i=0;i<n;i++)
    {
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    avg_wt /= n;
    avg_tat /= n;

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
    {
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
    }

    printf("\nAverage Waiting Time: %.2f",avg_wt);
    printf("\nAverage Turnaround Time: %.2f\n",avg_tat);

    return 0;
}
```

```
Enter the number of processes: 4
Enter arrival time and burst time for process 1: 1 2
Enter arrival time and burst time for process 2: 2 4
Enter arrival time and burst time for process 3: 4 6
Enter arrival time and burst time for process 4: 7 8

PID     AT      BT      CT      TAT     WT
1       1       2       3       2       0
2       2       4       7       5       1
3       4       6       13      9       3
4       7       8       21      14      6

Average Waiting Time: 2.50
Average Turnaround Time: 7.50

Process returned 0 (0x0)   execution time : 14.740 s
Press any key to continue.
```